2. Discuss the nature of solutions of the following system of equations

(i) $x+2y-z=6\,;\ -3x-2y+5z=-12\,;\ x-2z=3$

(ii) $2y+z=3\big(-x+1\big)\,;\ -x+3y-z=-4\,;\ 3x+2y+z=-\frac{1}{2}$

(iii) $\frac{y+z}{4}=\frac{z+x}{3}=\frac{x+y}{2}\,;\ x+y+z=27$

3. Vani, her father and her grand father have an average age of 53. One-half of her grand father's age plus one-third of her father's age plus one fourth of Vani's age is 65. Four years ago if Vani's grandfather was four times as old as Vani then how old are they all now ?

4. The sum of the digits of a three-digit number is 11. If the digits are reversed, the new number is 46 more than five times the former number. If the hundreds digit plus twice the tens digit is equal to the units digit, then find the original three digit number ?

5. There are 12 pieces of five, ten and twenty rupee currencies whose total value is ₹105. When first 2 sorts are interchanged in their numbers its value will be increased by ₹20. Find the number of currencies in each sort.

3.3 GCD and LCM of Polynomials

3.3.1 Greatest Common Divisor (GCD) or Highest Common Factor (HCF) of Polynomials

In our previous class we have learnt how to find the GCD (HCF) of second degree and third degree expressions by the method of factorization. Now we shall learn how to find the GCD of the given polynomials by the method of long division.

As discussed in Chapter 2, (Numbers and Sequences) to find GCD of two positive integers using Euclidean Algorithm, similar techniques can be employed for two given polynomials also.

The following procedure gives a systematic way of finding Greatest Common Divisor of two given polynomials $f(x)$ and $g(x)$.

Step 1: First, divide $f(x)$ by $g(x)$ to obtain $f(x)=g(x)q(x)+r(x)$ where $q(x)$ is the quotient and $r(x)$ is the remainder. Then, $\deg\big[r(x)\big]<\deg\big[g(x)\big]$

Step 2: If the remainder $r(x)$ is non-zero, divide $g(x)$ by $r(x)$ to obtain $g(x)=r(x)q_1(x)+r_1(x)$ where $r_1(x)$ is the new remainder. Then $\deg\big[r_1(x)\big]<\deg\big[r(x)\big]$. If the remainder $r_1(x)$ is zero, then $r(x)$ is the required GCD.

Step 3: If $r_1(x)$ is non-zero, then continue the process until we get zero as remainder. The divisor at this stage will be the required GCD.

We write $GCD\big[f(x),g(x)\big]$ to denote the GCD of the polynomials $f(x),g(x)$.

Note

If $f(x)$ and $g(x)$ are two polynomials of same degree then the polynomial carrying the highest coefficient will be the dividend. In case, if both have the same coefficient then compare the next least degree's coefficient and proceed with the division.

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